

A Touch Controller Using Differential Sensing Method for On-Cell Capacitive Touch Screen Panel Systems

Ik-Seok Yang, *Student Member, IEEE*, and Oh-Kyong Kwon, *Member, IEEE*

Abstract — A touch controller is proposed for on-cell capacitive touch screen panel systems. The proposed IC adopts the differential sensing method to enhance the dynamic range of sensing voltage and to be robust to display noise. The measurement results show that the maximum reporting rate, jitter tolerance, and signal-to-noise ratio (SNR) are 140 Hz, ± 0.3 mm, and 12 dB, respectively, when evaluated with a 13.3-inch wide extended graphics array (WXGA) liquid crystal display (LCD) panel with the on-cell touch screen. The proposed IC fabricated via 0.35 μ m CMOS process technology occupies a silicon area of 4 mm $\times$ 5 mm and consumes a power of 19 mW when the supply voltage is 3.3 V.¹

Index Terms — On-cell capacitive touch screen panel, mutual capacitance, differential sensing method, readout circuit.

I. INTRODUCTION

Touch screen panel (TSP) technology has come into the spotlight in information display systems because it provides a comfortable and intuitive user-interface (UI) and has advantages of easier and faster entry of information.

Many TSPs have been researched and commercialized to satisfy the demands related to the UI and information entry. Currently, three types of TSPs can be identified based on the location of the touch sensors: add-on types, on-cell types, and in-cell types. The touch sensors of add-on TSPs are the most common commercial type and are found in small to large panels, such as mobile phones, tablet PCs, and TVs, and are separated from the display panel. The manufacturing process for this TSP type has been refined, and many sensing methods have been reported [1]–[3]. In on-cell type TSPs, the touch sensors are located between the polarizer and the indium tin oxide (ITO) electrode in the display panel to make the display module slimmer [4], [5]. This type mainly exploits the capacitive sensing method for small and medium-sized panel applications such as mobile phones and mobile devices. Touch sensors in the in-cell type TSPs are located in the pixels. In-cell type TSPs are currently being intensively researched so that they can be applied to optical [6]–[8] and capacitive sensing methods [9]. However, these TSPs have not yet been

commercialized due to a number of processing and reliability issues.

Many sensing methods for add-on TSP applications, such as the resistive [1], capacitive [2], [3], acoustic-wave [10], and infrared methods [11], have been commercialized. Among them, capacitive touch screens have been widely adopted in high-end mobile applications such as smart phones and tablet PCs because they provide not only multi- and soft-touch features but also higher durability and superior light transmittance over a resistive touch screen. Furthermore, most display module makers have tried to adopt the on-cell capacitive TSP, which enables a reduction in the system cost and the manufacture of small and slim devices by merging the TSP with the display panel [4], [5].

In the on-cell capacitive TSP, the touch sensor is located closely to the common electrode of display panels as previously mentioned. The capacitive coupling between TSP electrodes and display panel electrodes during display operation is very significant compared to an add-on capacitive TSP. This makes it difficult to detect the exact touch position due to sensing errors generated by this coupling noise during display operation. In particular, as the size of the display panel increases, the capacitive coupling noise between the display panel and the TSP becomes more serious. In addition, the variation in capacitance due to a touch event is so small, so that it is difficult to detect. This results in a lower signal-to-noise ratio (SNR), which eventually prevents detection of the exact touch position. Therefore, a new sensing method is necessary in order to convert small capacitance variation to voltage and to enhance the dynamic range of the sensing voltage without distortion.

In this paper, a touch controller using a differential sensing method is proposed to enhance the dynamic range of the sensing voltage and to reduce sensing errors due to coupling noise during display operation.

II. ON-CELL CAPACITIVE TOUCH SCREEN PANEL SYSTEMS

A. On-cell Capacitive Touch Display Systems

A conventional liquid crystal display (LCD) system with add-on capacitive TSP is shown in Fig. 1(a). It is composed of four parts: a glass, an add-on touch panel, a polarizer, and a display panel. The add-on touch panel is separated from the

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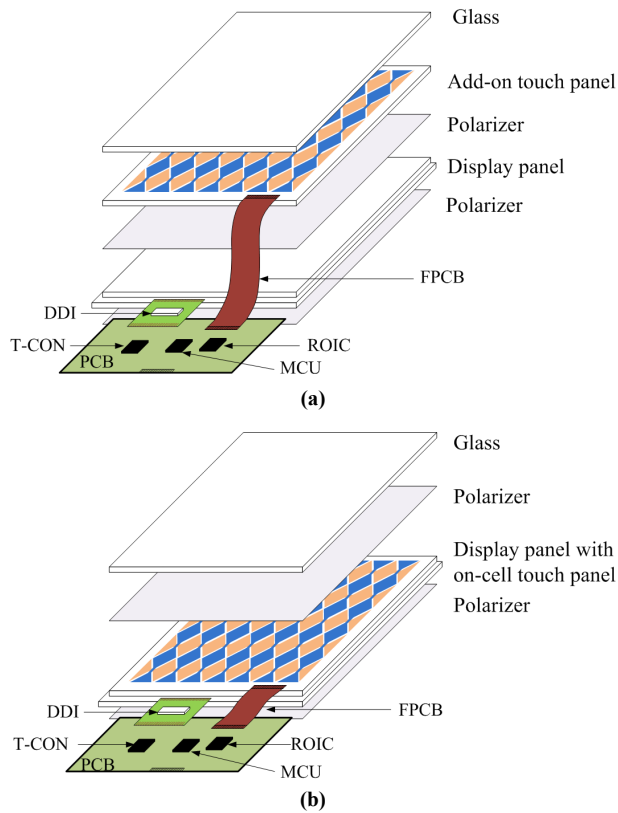


Fig. 1. Configurations of a LCD system with (a) an add-on capacitive TSP and (b) an on-cell capacitive TSP.

display panel. A display driver IC (DDI) is connected between the display panel and the timing controller IC (T-CON) in the printed circuit board (PCB). Also, the readout IC (ROIC) and the microcontroller unit (MCU) are mounted on the PCB. The ROIC is linked to the sensor electrodes via a flexible printed circuit board (FPCB). The ROIC and MCU can be merged onto one chip as is the case with TSP applications [5], [9].

Fig. 1(b) shows an LCD system with an on-cell capacitive TSP. It is composed of three parts: a glass, a polarizer, and an on-cell touch display panel of wherein the surface is patterned by transparent electrode materials. As shown in Fig. 1, the manufacturing process for on-cell type touch displays is the same that for an add-on type touch displays except for the ITO patterning on the display panel. Therefore, the on-cell type touch display reduces the module-assembly cost while it enables slimmer module design with better display quality.

B. Mutual Capacitance Sensing Method for Multi-touch Recognition

The readout circuit can detect a touch position by measuring the capacitance changed due to touch event. The self capacitance sensing method [5] and mutual capacitance sensing method [12] are primarily used to measure capacitance variation. Fig. 2 illustrates the self capacitance sensing method, which measures the self capacitance of each TSP lines and detects the touch position. When a touch object is touched, the capacitance of the TSP lines close to the touch object increases. The touch position can be then detected according to a capacitance profile with respect to TSP lines.

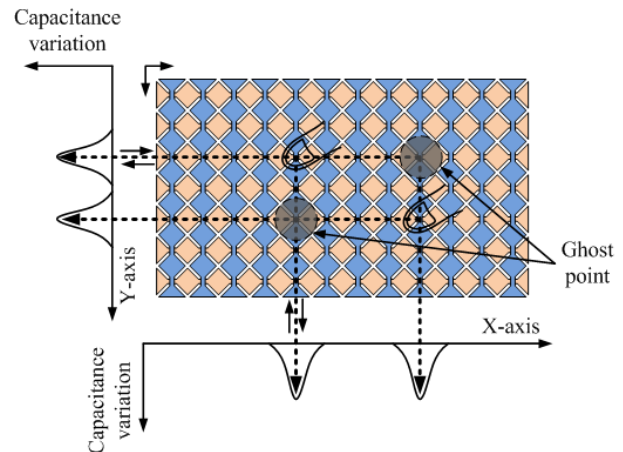


Fig. 2. Ghost point failure when a self capacitance sensing method is adopted in capacitive TSP.

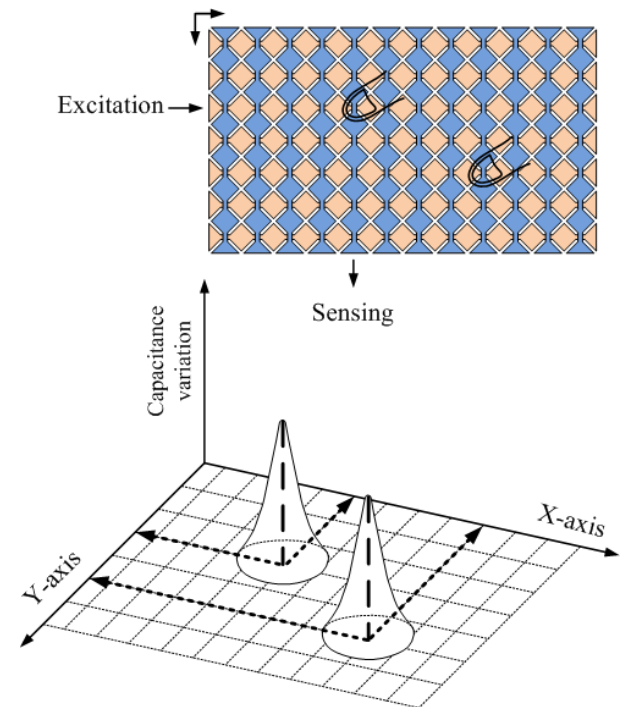


Fig. 3. Mutual capacitance sensing method in capacitive TSP.

However, a multi-touch event cannot be recognized with this method because the untouched positions (called ghost points) can be detected simultaneously as shown in Fig. 2.

Fig. 3 shows the mutual capacitance sensing method, which measures the mutual capacitance of all sensors in a TSP compared to the self capacitance sensing method. When a touch object is touched, the mutual capacitance formed between the driving lines and sensing lines decreases. The touch position can be detected using the capacitance profile with respect to all touch sensors. Therefore, the touch position can be identified without error and a multi-touch event can be recognized using this method.

C. Technical Considerations of the On-cell Capacitive Touch Display Systems

Although the on-cell type touch display system has many advantages over the add-on type, there are a few technical challenges to overcome.

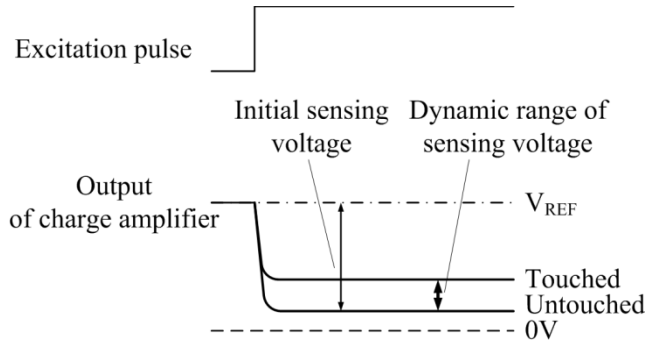


Fig. 4. Dynamic range of the sensing voltage in a charge amplifier when a conventional single excitation pulse is used for a mutual capacitance sensing method.

One challenge is to reduce display noise during display operation. The TSP lines are located closely to the display panel [5]. This causes a large vertical parasitic capacitance between the TSP lines and a common electrode in the display panel, resulting in increased capacitive coupling of display noise to the sensing circuit and a poor SNR.

Another technical challenge is sensing a change in the smaller signal component due to a touch event from the larger base component due to the initial mutual capacitance. Fig. 4 shows the signal dynamic range according to the base component, which is called the initial mutual capacitance. The charge amplifier converts the small signal component as well as the large component to a voltage [12]. Therefore, this larger base component results in a reduced dynamic range allocated to the signal component and restricts the amplification of this range due to signal saturation, causing a narrow input analog-to-digital conversion (ADC) range and poor sensor sensitivity.

III. TOUCH CONTROLLER WITH THE PROPOSED SENSING METHOD

A. Proposed Differential Sensing Method

Fig. 5 shows a schematic and a timing diagram of the proposed sensing circuit. It is composed of a differential amplifier, integration capacitors, reset switches, and switches for polarity inversion. The mutual capacitance sensing method is used for multi-touch recognition. To achieve robustness to noise and the absolute value of a mutual capacitance, complementary excitation pulses are applied to two adjacent driving lines and the mutual capacitance between the driving lines and sensing line is sensed. In Fig. 5(a), $R_{t,n-1}$ and $C_{t,n-1}$ are the resistance and stray capacitance of the (n-1)-th driving line on TSP, respectively, and $R_{t,n}$ and $C_{t,n}$ are the resistance and stray capacitance of the n-th driving line on the TSP, respectively. $R_{r,m}$ and $C_{r,m}$ are the resistance and stray capacitance of the m-th sensing line on the TSP, respectively, and $C_{s,n-1,m}$ and $C_{s,n,m}$ is the mutual capacitance between the m-th sensing line and the (n-1)-th driving line and n-th driving line on the TSP, respectively.

In the reset phase, the RST signal is high, the integration capacitor is discharged and differential outputs of an amplifier

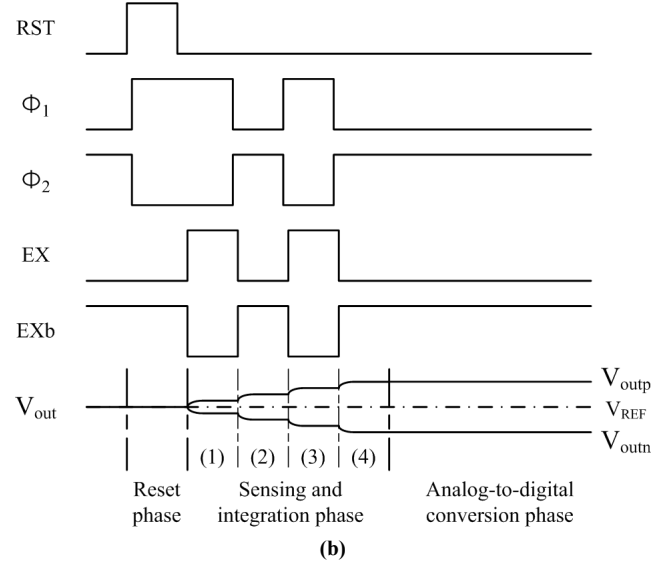
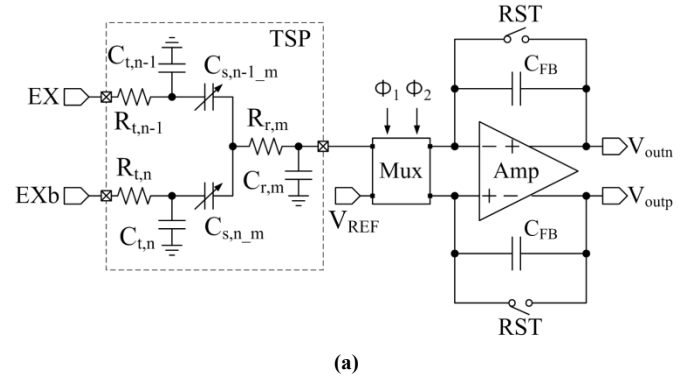


Fig. 5. (a) Schematic diagram of the proposed sensing circuit and (b) its timing diagram.

are initialized at V_{REF} . In the sensing and integration phase, the complementary excitation pulse is applied to the two adjacent driving lines on the TSP and the input voltage is connected to an amplifier via a multiplexing switch by controlling the polarity of the input stage in the amplifier. The amplifier and the integration capacitor serve as a charge amplifier; that is, they convert the charge induced by excitation pulses to a voltage. At that time, the differential output voltage of the amplifier is determined as follows:

$$V_{outn} = V_{REF} - \frac{C_{s,n-1,m} - C_{s,n,m}}{C_{FB}} \times \Delta V_{EX} \quad (1)$$

$$V_{outp} = V_{REF} + \frac{C_{s,n-1,m} - C_{s,n,m}}{C_{FB}} \times \Delta V_{EX} \quad (2)$$

$$V_{out} = V_{outp} - V_{outn} = 2 \cdot \frac{C_{s,n-1,m} - C_{s,n,m}}{C_{FB}} \times \Delta V_{EX} \quad (3)$$

where V_{outp} and V_{outn} are the positive and negative output voltages of the amplifier, respectively, V_{out} is the differential output voltage of the amplifier, and ΔV_{EX} is the swing voltage of the complementary excitation pulse. As shown in (1) and (2), the differential output of the amplifier is expressed as the difference between two mutual capacitances between two adjacent driving lines and the sensing line by eliminating the

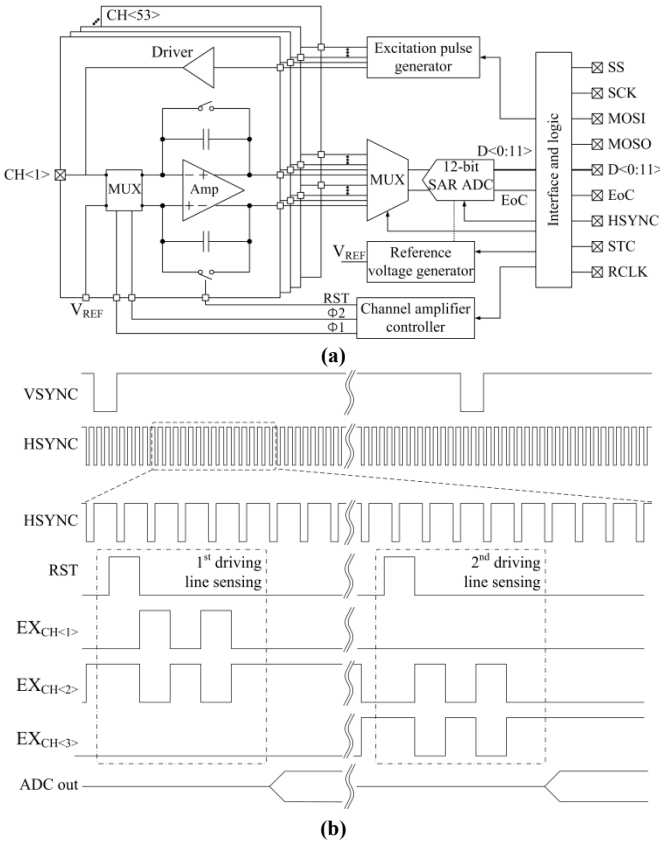


Fig. 6. (a) Block diagram of readout IC with the proposed sensing method and (b) its overall timing diagram.

absolute value of the mutual capacitance. Therefore, a wide dynamic range of differential amplifier outputs can be obtained. Thus, the proposed sensing method can be adopted in most capacitive TSPs independent of the size and resolution. Through four repeated sensing and integration steps, V_{out} in (3) is increased four-fold. The proposed sensing method also reduces the display noise and improves SNR. In the ADC phase, sensing voltage is converted to digital data.

B. Readout IC with the Proposed Sensing Method

Fig. 6(a) shows a block diagram of readout IC with the proposed sensing method. It is composed of 53-sensing channels, 53-excitation pulse generators, a multiplexer, a 12-bit successive approximation register (SAR) ADC [13], a reference voltage generator, a control logic block, and a serial peripheral interface (SPI) block. As shown in Fig. 6(a), the sensing circuit and excitation driver are integrated into a channel, such that one channel selectively serves as either the sensing circuit or excitation driver.

Fig. 6(b) shows the overall timing diagram of the proposed readout IC when the complementary excitation pulses are applied to two adjacent driving lines.

The excitation pulses are designed to be synchronous with the display driving signals such as VSYNC and HSYNC in order to minimize the display noise. Sensing operations of all channels are performed simultaneously and stored in the sensing voltage as (1) and (2). After the sensing operation, these voltage is sequentially sampled by MUX and converted

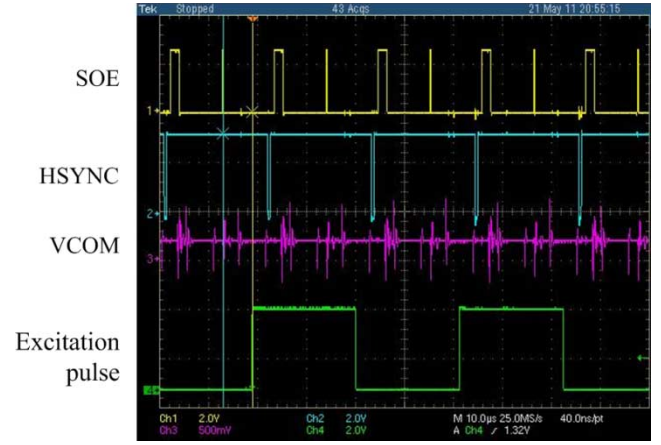


Fig. 7. Waveform of the VCOM electrode measured during display operation and timing of and excitation pulse. The horizontal scale is 10 μs/div for all signals and the vertical scale is 2 V/div for all signals except VCOM, which is 500 mV/div.

to digital data by ADC. This process is repeated until the last driving line is excited and then the $n \times m$ -pattern image is formed and stored in MCU. By image processing using the $n \times m$ -pattern image, the MCU extracts the exact touch coordinates.

C. Minimizing the Display Noise Effect

In the display system with the on-cell capacitive TSP, the coupling noise between the TSP electrodes and the VCOM electrode during display operation is very critical because they are located close together. Fig. 7 shows a noise waveform of the VCOM electrode in the LCD system during display operation. SOE and HSYNC are the output driving enable signals of the data and gate driver, respectively. The VCOM electrode fluctuates after these signals falling edge. In order to minimize the display noise effect, the excitation pulse must be designed to be synchronous with display driving signals such as VSYNC and HSYNC such that the sensing operation is performed after display driving signals have stabilized as shown in Fig. 7. By synchronizing the timing of the sensing circuit with display driving timing, the sensitivity and display noise immunity can be improved.

IV. EXPERIMENT RESULTS

Fig. 8 shows the display system with an on-cell capacitive TSP with the proposed method. The display system consists of a TSP readout board composed of two readout ICs for driving and sensing and a MCU, a display driving board, and a display panel with an on-cell capacitive TSP. The structure of the TSP includes 53×29 diamond-type ITO electrodes, and the pitch of the ITO lines is 5.5 mm. The specifications of the display system are shown in TABLE I. Information regarding the sensing capacitance of the TSP was stored in the memory of MCU. Measurement results show that the mutual sensing capacitances of the TSP were 2.2 and 1.85 pF when TSP was touched and untouched, respectively.

Fig. 9 shows the measurement results of mutual capacitances on the TSP when five 5-mm-diameter touch objects are used. Fig. 9(a) and 9(b) show images of ADC data

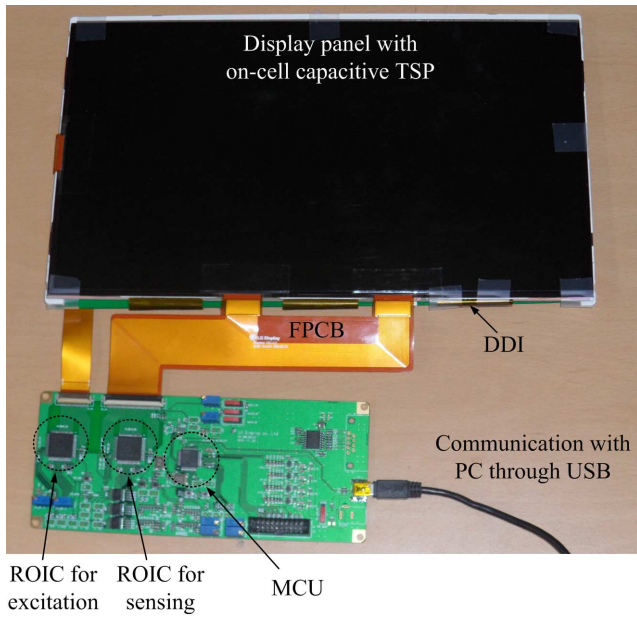


Fig. 8. Display system with an on-cell capacitive TSP. For the experiment, the touch control board was separated from the display driving board, which is located on the backside of display panel.

from measured mutual sensing capacitances. These data indicate that there are five touch positions. Also, it is evident that the sensing operation works well even though the swing voltage of excitation pulses is 3.3 V due to a wide dynamic range of sensing voltages. Fig. 9(c) shows output waveforms of the proposed sensing circuit when the third driving line is excited. The maximum differential value appeared at the seventh sensing line as position “A” shown in the 2-D image in Fig. 9(a).

TABLE II summarizes the performance of the proposed readout IC. Specially, the table lists the following touch performance parameters: maximum 140-Hz reporting rate, ± 0.3 -mm jitter tolerance, 12-dB SNR, and reflects the display performances of the 13.3-inch LCD system in wide extended graphics array (WXGA) resolution format. The power consumption of the proposed readout IC is 19 mW when the supply voltage is 3.3 V.

TABLE I
SPECIFICATION SUMMARY FOR THE PROPOSED TSP SYSTEM

Properties		Specifications
TSP	Sensor type	On-cell capacitive
	ITO pattern type	Diamond
	Pattern pitch	5.5 mm
	Spatial resolution	53×29
	Diagonal size	13.3-inch
Sensing block	Sensing method	Mutual capacitance
	Number of channels	53
ADC	Type	SAR ADC
	Resolution	12-bit
	Sampling rate	600 k sample/s
Processor		ARM

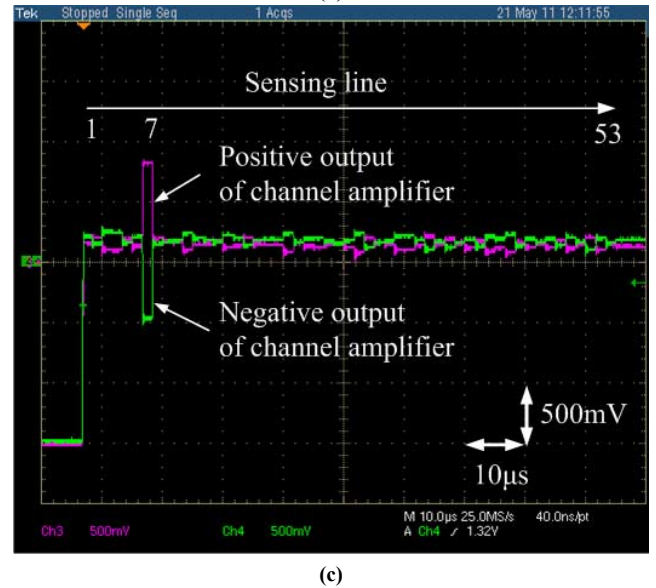
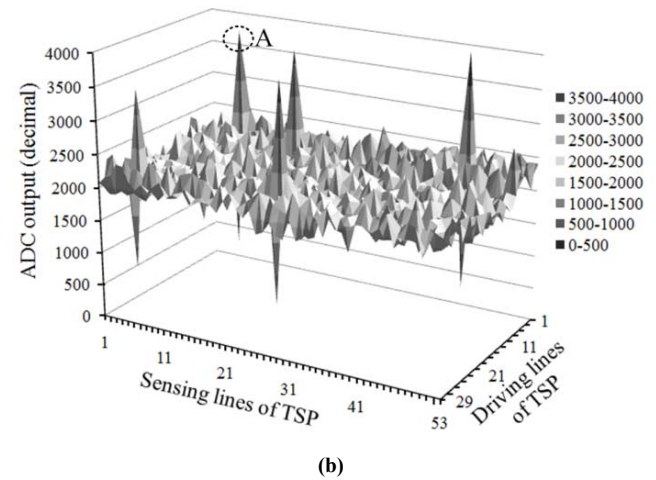
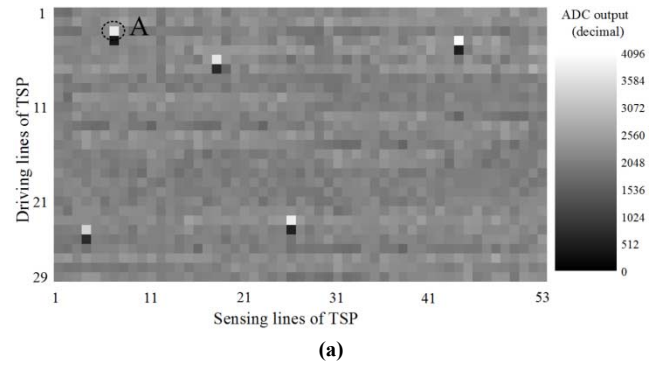


Fig. 9. (a) 2-D and (b) 3-D image based on the measurement results of mutual capacitances when the TSP resolution is 53×29 and 5 touch objects with 5-mm diameter are touched. (c) Measured output waveforms of the proposed sensing circuit when the third driving line is excited. The maximum differential value appeared at the seventh sensing line as shown in a 2-D image of (a).

V. CONCLUSION

A touch controller was proposed for on-cell capacitive TSP systems. The proposed IC adopts the differential sensing method to enhance the dynamic range of the sensing voltage and to provide robustness to display noise. The proposed IC exhibits a maximum touch performance of a 140 Hz reporting

rate, ± 0.3 mm jitter tolerance, and a 12 dB SNR when the supply voltage is 3.3 V and the swing voltage of excitation voltage is 3.3 V and evaluated using 13.3-inch WXGA LCD panel with an on-cell touch screen. The proposed IC fabricated via 0.35 μm CMOS process technology consumes a power of 19 mW when the supply voltage is 3.3V. The proposed method can also recognize a multi-touch event. Therefore, the proposed readout IC can be available in the medium-sized on-cell capacitive TSPs.

TABLE II
PERFORMANCE SUMMARY FOR THE PROPOSED READOUT IC

Properties	Specification
Swing voltage of excitation pulse (V)	3.3
Maximum reporting rate (Hz)	140
SNR (dB)	12.6
Jitter (mm)	± 0.3
Power consumption (mW)	19

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BIOGRAPHIES



Ik-Seok Yang (S'07) received a B.S. and M.S. in Electronics Engineering from Hanyang University in Seoul, Korea in 1996 and 1998, respectively. From 1998 to 2006, he was with Magnachip Semiconductor Inc., in Seoul, Korea, where he was involved in the development of process integration and driving circuit design for flat panel display applications. He is currently pursuing a Ph. D. at Hanyang University in Seoul, Korea. He has been engaged in research on sensing methodologies for touch screen panel systems.

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